Exam 2.E · Blank Mock Test

Name: _____ Section: 12 / 29

Practice / Class Review. Show all work. This blank copy corresponds to the uploaded annotated Exam 2.E solutions.

1. Factor completely: $25x^2 - 36y^2$.

2. Solve $-13 < 3x - 1 \leq 5$. Write interval notation and graph on a number line.

Use a separate sheet of graph paper or draw your graph below.

3. Solve $-9x + 5 < -3(x - 7)$. Write the answer in interval notation.

4. Multiply and simplify: $(x + 4)(x^2 - 2x + 3)$.

5. Solve $3|x - 1| + 2 \leq 14$. Give the answer as an interval.

6. Factor completely: $2x^3 + 128$.

7. Factor by grouping: $x^2y + 7y + 4x^2 + 28$.

8. Solve $(5x - 2)/(-4) \geq 7$. Write interval notation.

9. Graph the system: $x + 2y \geq 2$ and $3x - y < 6$.

Use a separate sheet of graph paper or draw your graph below.

10. Solve $3x + 7 \leq x - 1$ OR $5x - 6 > 2x + 9$. Write interval notation and graph.

Use a separate sheet of graph paper or draw your graph below.

11. Solve $p(p + 3) = 54$.

12. Solve $2|3x - 5| + 7 = 25$.

13. Factor completely: $18x^2 - 39x + 18$.

14. Factor completely: $3t\blacksquare - 6t\blacksquare - 45t\blacksquare$.

15. Subtract and simplify: $(7x\blacksquare - 3x^2 + 4x - 9) - (2x\blacksquare + 5x^3 - 8x^2 - 6)$.

16. Graph the inequality $y \leq -(1/2)x + 3$.

Use a separate sheet of graph paper or draw your graph below.

17. The product of two consecutive page numbers is 306. Find the page numbers.